

SIMANTA SARKAR

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FPGA/RTL Design Engineer with hands-on experience across JESD204C RF data-converter bring-up (AD9082, AFE7950), AXI DMA architectures, RTL verification (Vivado XSIM), and embedded Linux on Xilinx Zynq/Zynq UltraScale+ MPSoC platforms.

EDUCATION

Indian Institute of Information Technology Guwahati — Guwahati, Assam

July 2024 – July 2026

Master of Technology in VLSI and Embedded Systems | **CGPA: 8.08**

Jorhat Institute of Science & Technology — Jorhat, Assam

Aug 2020 – July 2024

Bachelor of Technology in Electrical, Electronics and Communications Engineering | **CGPA: 6.92**

TECHNICAL SKILLS

RTL & Verification: Verilog, VHDL, RTL Design, FSM Architecture, Testbench Development, Functional Simulation (Vivado XSIM), Timing Closure, ILA/ChipScope Hardware Debug

High-Speed Protocols: JESD204B/C, AXI4, AXI4-Stream, AXI4-Lite, AXI DMA (Scatter-Gather), SPI, UART, Ethernet (lwIP/UDP), PCIe

FPGA/SoC Platforms: Xilinx Zynq-7000, Zynq UltraScale+ MPSoC, KCU105, VMK180, ZC702, ZCU102, ZCU104, ZedBoard, Artix-7

Data Converters & Mixed-Signal: LTC2175, AD9082, AFE7950, Multi-GSPS ADC/DAC Interfacing, Clock Synthesis (HMC7044), PLL/MMCM

Embedded & Tools: PetaLinux, Device Tree, Vitis, Embedded C, Python, C++, Xilinx Vivado, MATLAB/Simulink

PROFESSIONAL EXPERIENCE

FPGA Engineer | Berylline Labs Pvt. Ltd. — Kolkata, WB

Feb 2026 – Present

- Bring up JESD204C RF data-converter links for AD9082 (VMK180) and AFE7950 (KCU105, ZCU102), validating 4 GSPS symmetric and 6/12 GSPS asymmetric lane configurations at line rates up to 24.75 Gbps.
- Build and debug PetaLinux embedded Linux images for Zynq UltraScale+ MPSoC targets — resolving device-tree corruption and duplicate phandle conflicts, and correcting an HMC7044 PLL mis-lock that was blocking JESD204C FSM bring-up.
- Architect AXI4-Stream and Scatter-Gather DMA pipelines for multi-channel ADC/DAC data movement between PL and PS, enabling real-time RF signal acquisition streamed to a host over Ethernet; root-cause hardware defects via ILA-based waveform capture, including clock-domain routing and datapath width-mismatch issues.
- Contribute to additional cross-domain deliverables: a VHDL sensor-fusion LiDAR system for railway pole detection (CSIR-CGCRI-sponsored, ZCU104/Arty A7, UART packet-parser FSM), a multi-rate DSP filter chain (51-tap Kaiser FIR, >80 dB stopband attenuation), and a PLL/MMCM-driven ultrasound carrier design with safety-FSM and SPI DAC control for a medical stimulator device.

Signal & Telecom Intern | Indian Railways — Guwahati, Assam

Jul 2023 – Aug 2023

- Audited 50+ railway signaling network nodes and tested interlocking/track-circuit monitoring equipment, contributing to a 20% increase in network throughput.

Python Development Intern | IIT Guwahati — Guwahati, Assam

Aug 2022 – Oct 2022

- Engineered Python automation scripts for LaTeX corpus analysis (100% extraction accuracy) and built a Tkinter GUI that cut manual processing time by one-fifth.

Deep Learning Research Intern | IIT Guwahati — Guwahati, Assam

Aug 2022 – Sep 2022

- Extended a Deep-Waveform learned OFDM receiver architecture under Prof. Dr. Kuntal Deka, restructuring the DNN to cut symbol classification error by 5% across SNR sweeps.

ACADEMIC PROJECT

Multi-Channel FPGA ADC Data Acquisition System (M.Tech Thesis) | Zynq-7020, LTC2175, AXI DMA, Gigabit Ethernet

- Designed a 4-channel, 14-bit real-time acquisition pipeline (LTC2175 ADC → AXI-Stream FIFO → Scatter-Gather DMA → lwIP UDP → PyQt6 GUI); resolved a DMA-idle FIFO-overflow bug and a double-buffer cache-coherency race.
- Implemented a centroid-based peak-detection FSM for FBG interrogation and used ILA to root-cause a clock-routing defect (missing BUFG) that was stalling waveform capture.
- Wrote Verilog/VHDL testbenches and ran functional simulation in Vivado Simulator (XSIM) to validate DMA control logic and peak-detection FSM behavior prior to hardware bring-up.

ADDITIONAL PROJECTS

Memristor-Inspired Spiking Neuromorphic Architecture for Hand Gesture Recognition — Hybrid neuromorphic system combining MediaPipe landmarks with a memristor-driven SNN; 99.8% accuracy on 21-class keypoint classification at 20+ FPS for rehabilitation applications.

Diabetic Retinopathy Detection with Deep Learning — AI classification pipeline reaching 83% accuracy; findings presented at the International Conference on Contemporary Researches in Engineering, Science, Management & Arts.

Embedded Indicators Control using PSoC and Simulink — Real-time automotive indicator control system using MATLAB Simulink Stateflow and PSoC Creator 4.4, integrating FSM logic across GPIO, PWM, Timer, and UART modules.

Automated Number Plate Detection (ANPR) — OpenCV-based detection pipeline achieving 89% accuracy across 1,000+ images and 20+ hours of video.

CERTIFICATIONS

MATLAB Certified | Data Science with Python — UDGAM Program, IIT Guwahati